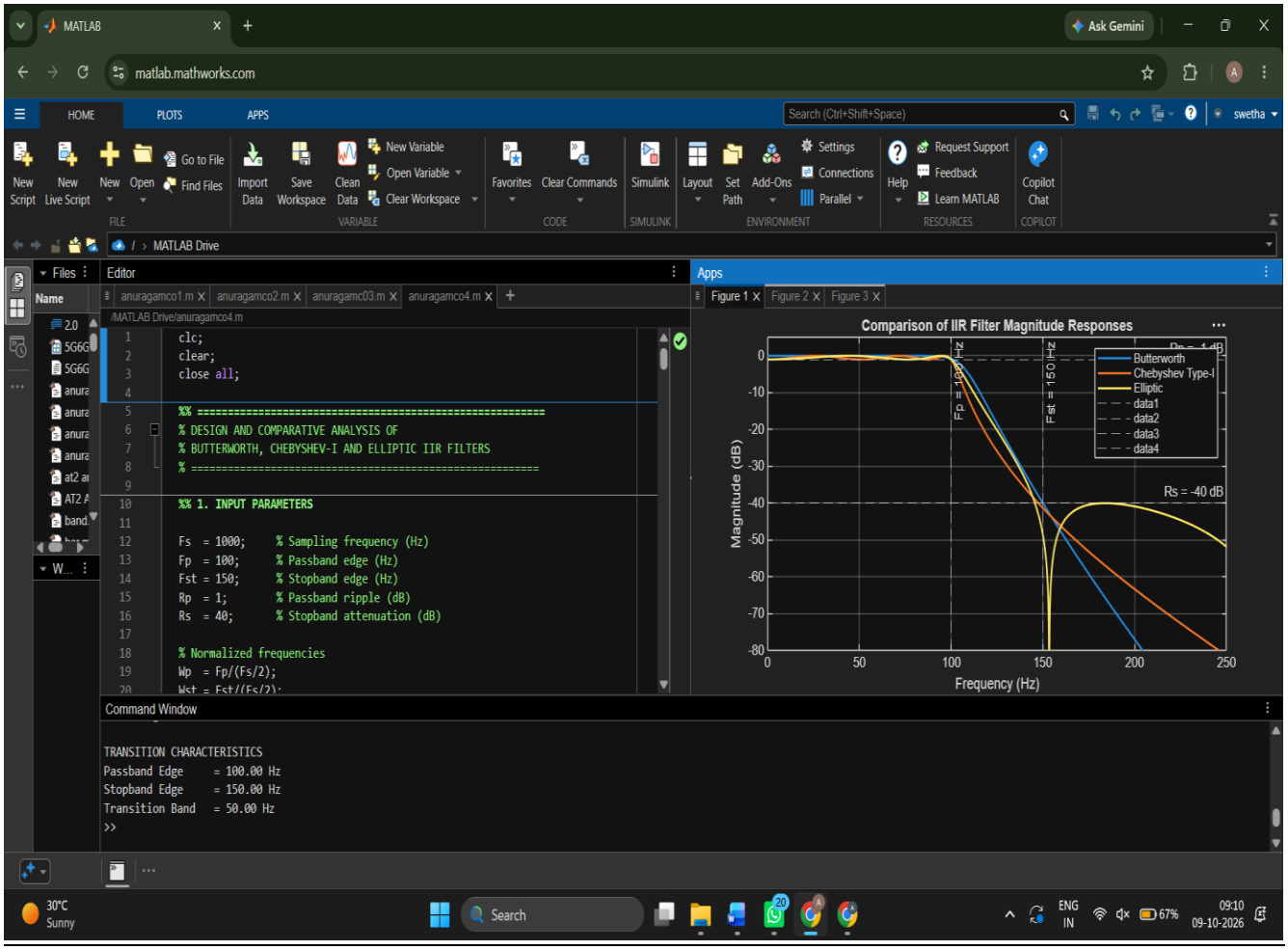
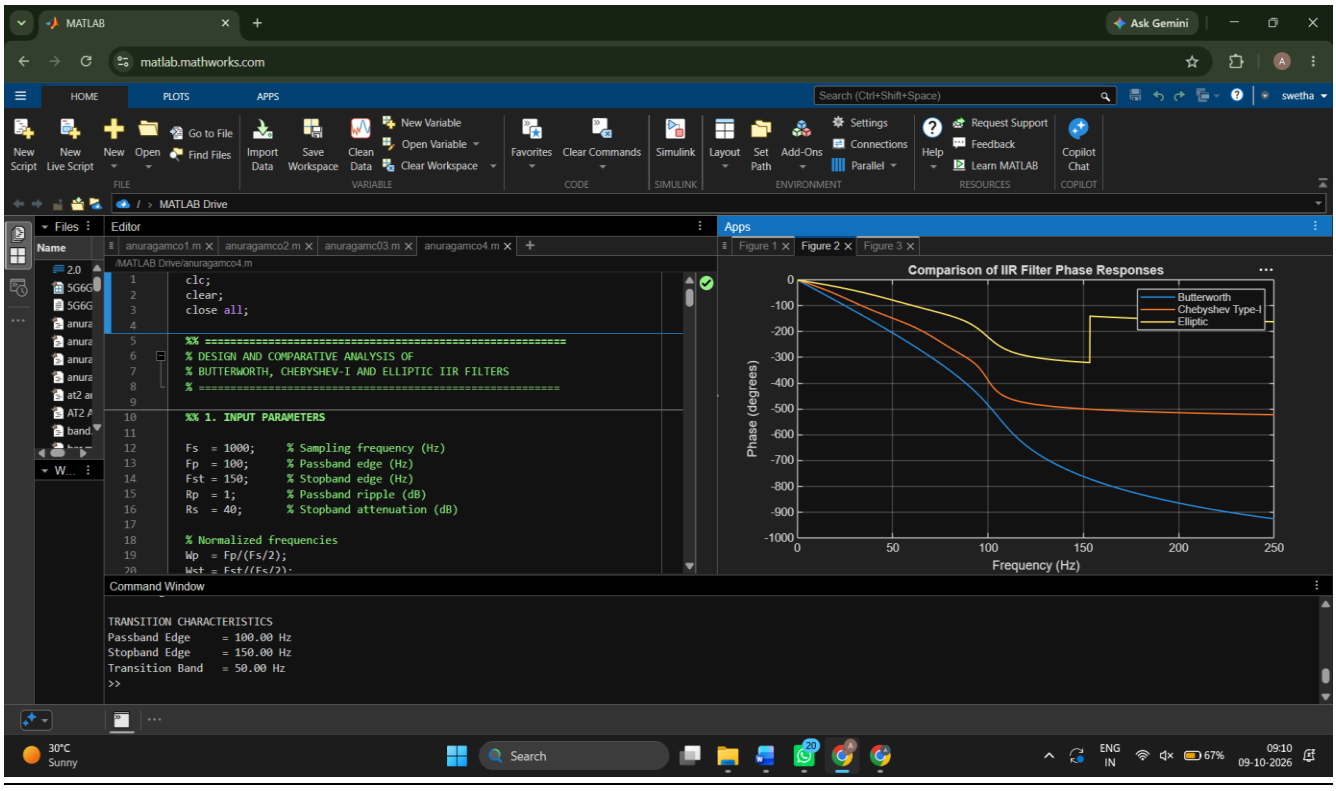
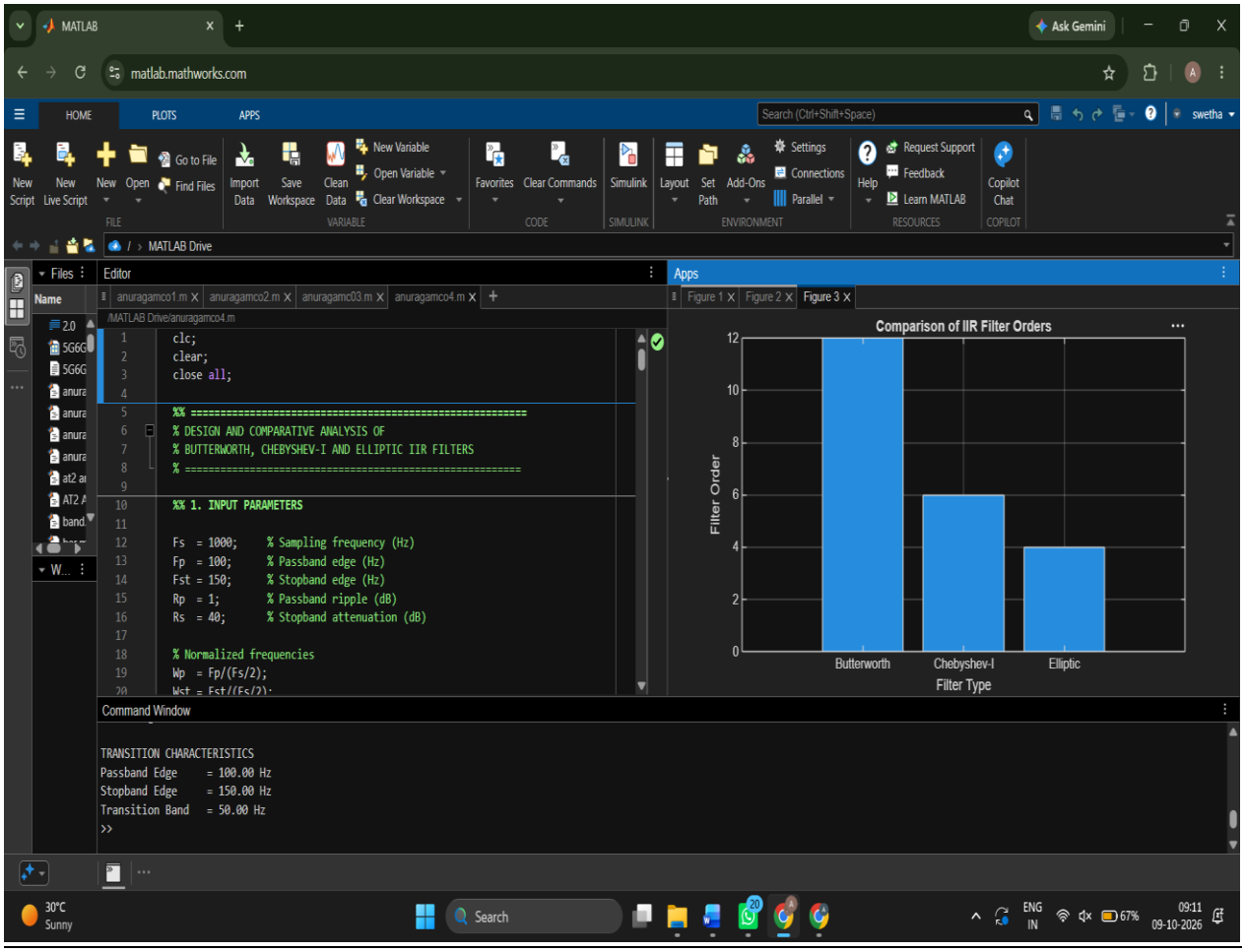

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Name
Editor
anuragamco1.m
anuragamco2.m
anuragamco03.m
anuragamco4.m
/MATLAB Drive/anuragamco4.m
clc;
clear;
close all;
%% ============================================================
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ============================================================
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
% Normalized frequencies
Wp = Fp/(Fs/2);
Apps
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Magnitude Responses
Butterworth
Chebyshev Type-I
Elliptic
data1
data2
data3
data4
Rs = -40 dB
Magnitude (dB)
Frequency (Hz)
Command Window
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
30°C
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Open
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Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
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Set Path
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/ › MATLAB Drive
Files
Name
Editor
anuragamco1.m
anuragamco2.m
anuragamco03.m
anuragamco4.m
/MATLAB Drive/anuragamco4.m
clc;
clear;
close all;
%% ============================================================
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ============================================================
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
% Normalized frequencies
Wp = Fp/(Fs/2);
Apps
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Phase Responses
Butterworth
Chebyshev Type-I
Elliptic
Phase (degrees)
Frequency (Hz)
Command Window
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
30°C
Sunny
Search
ENG
IN
67%
09:10
09-10-2026


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Ask Gemini
HOME
PLOTS
APPS
Search (Ctrl+Shift+Space)
swetha
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
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anuragamco4.m
/MATLAB Drive/anuragamco4.m
Figure 1
Figure 2
Figure 3
clc;
clear;
close all;
%% ============================================================
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ============================================================
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
% Normalized frequencies
Wp = Fp/(Fs/2);
Comparison of IIR Filter Orders
Filter Order
Filter Type
Butterworth
Chebyshev-I
Elliptic
0
2
4
6
8
10
12
Command Window
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>
30°C
Sunny
Search
ENG
IN
67%
09:11
09-10-2026